

# Introduction to Finance

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## 1. Lec 3. Asset Classes

### 1.1. Equities

Common Stock

- Ownership in a company
- Voting rights at shareholder meetings, residual claim on assets
- Dividends are not guaranteed

$$\text{Market Value} = \text{Number of Shares} \times \text{Price per Share}$$

### 1.2. Derivatives

#### 1.2.1. Option

**Definition 1.2.1.1 Option**

**Call Option** gives the holder the right to buy an asset at a specified (exercise) price on or before a specified expiration date.

**Put Option** gives the holder ...

Option confers a **right**, not an **obligation**.

**Definition 1.2.1.2 Future**

## 2. Lec. 4 Trading, Margin and Short Sale

### 2.1. How Firms Issue Security

#### 2.1.1. Primary Market: New Issues of Stocks(Bonds)

- Public vs. Private Corporations: The requirement of public disclosure, and liquidity of shares.
- **Initial Public Offerings (IPOs)**: the first issue of stocks to the public, where bulks of IPO's go to Institutional Investors. Investment bankers determine suitable prices, identify potential buyers, and (sometimes) commit to buy unsold shares.
- **Process of IPO**:
  - Underwriter: a financial institution that helps the firm issue securities.
  - Prospectus
  - Road Show

1. **Secondary Market**: Trading of Outstanding Securities (Securities already issued)

Organized secondary markets give assets liquidity.

## 2.2. How Securities are Traded

### 2.2.1. Types of Markets

- Direct Search: buyers and sellers search for each other.
- Brokered Markets: brokers search for buyers and sellers. (e.g. the primary market)
- Dealer Markets: dealers have inventories of assets from which they buy and sell.
- Auction Markets: all traders converge at one place to buy or sell.

### 2.2.2. Bid and Ask Prices

- The market maintains a limit order book:
- **Bid**: highest price at which someone is willing to buy;
- **Ask**: lowest ... sell
- **Spread**: difference between bid and ask. Ask must be higher than bid, otherwise the deal is done immediately.

### 2.2.3. Orders

#### Definition 2.2.3.1 Market Order

buy/sell at the best available price.

- Only quantity is specified, execution is immediate.
- Posted ask price may change before the order is executed or as a result of the order.

#### Note 2.2.3.1 Potential Problems with Market Orders

- Large Orders: may not be executed at the same price.
- Other Traders: another trader can beat our investor to the punch. Our order gets executed at a higher price.
- Sudden Price Changes: the price at which the order is executed may be different from the price at which the order was placed.

#### Definition 2.2.3.2 Limit Order

buy/sell specified quantity at a specified price.

- Limit order is one kind of the *price-contingent order* (the other being the stop order).
- All limit orders are stored in the limit order book, which serves as a source of bid and ask prices.

## 2.3. Margin Trading

### Definition 2.3.1 Buying on Margin

borrowing money to buy securities.

- **Debt-financed asset purchases:** some money put up by the investor, the rest borrowed from the broker.
- **Broker's Call:** the broker can make a margin call if the value of the account falls below a certain level.
- Shares are used as *collateral* for the loan.
- There is **Margin Requirement:** the minimum portion of the purchase price contributed by the investor.

### Definition 2.3.2 Margin Requirements

- **Initial Margin:** the minimum margin that must be supplied at the time of purchase, usually 50%.
- **Maintenance Margin:** applies subsequently, sensitive to the value of the account.

If price drop so far that the margin requirement is not met, the broker will issue a margin call.

Before giving an example on margin trading, we need to understand the **balance sheet** of a margin account.

### Theorem 2.3.1 Equity in a Margin Account

$$\text{Assets} = \text{Liabilities} + \text{Equity}$$

### Example 2.3.1 Margin Trading Example

Investor balance sheet (initially):

| Assets                | Liabilities | Equity |
|-----------------------|-------------|--------|
| Hydrochloric Acid     | 12.0        | 92.1   |
| Sodium Myreth Sulfate | 16.6        | 104    |
| Potassium Hydroxide   | 24.7        |        |

Suppose an investor buys 100 shares of a stock at \$50 per share, with an initial margin of 50%. The stock pays no dividends. The maintenance margin is 30%.

- **Initial Investment:** \$2500
- **Loan:** \$2500
- **Total Investment:** \$5000

If the stock price falls to \$40, the investor's equity is \$1500, which is below the maintenance margin of \$1200. The broker will issue a margin call.

The investor can either deposit more money or sell some shares to meet the margin requirement.

#### 2.3.1. Short Sales (Selling Short)

**Purpose:** profit from a decline in the price of a security.

**Mechanism:** borrow shares from a broker, sell them and deposit proceeds and margin in an account (avoiding the risk of default); buy them back later to return to the broker (covering the short position).